



## Practice assignment 9 - Not Graded



This assignment will not be graded and is only for practice.

### Multiple Select Question (MSQ)

1 point

Consider a function  $f: \mathbb{R}^2 \rightarrow \mathbb{R}$  defined as:

$$f(x, y) = \begin{cases} \frac{2xy}{\sqrt{x^2 + y^2}} & \text{if } (x, y) \neq (0, 0) \\ 1 & \text{if } (x, y) = (0, 0). \end{cases}$$



$$f(x, y) = \begin{cases} 2xy & \text{if } (x, y) \neq (0, 0) \\ 1 & \text{if } (x, y) = (0, 0). \end{cases}$$

Choose the set of correct options.



$$|x| \leq \sqrt{x^2 + y^2} \quad |x| \leq x^2 + y^2$$



$$\text{and } |y| \leq \sqrt{x^2 + y^2} \quad |y| \leq x^2 + y^2$$



If  $(x, y) \rightarrow (0, 0)$ , then  $f(x, y) \rightarrow 0$ .



$\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$  exists.



$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0$ .



$f(x, y)$  is continuous at the origin  $(0, 0)$ .

**No, the answer is incorrect.**

**Score: 0**

### Accepted Answers:

$$|x| \leq \sqrt{x^2 + y^2} \quad |x| \leq x^2 + y^2$$



$$\text{and } |y| \leq \sqrt{x^2 + y^2} \quad |y| \leq x^2 + y^2$$



If  $(x, y) \rightarrow (0, 0)$ , then  $f(x, y) \rightarrow 0$ .

$\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$  exists.

$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0$ .

1 point

Consider the following functions

$$f(x, y) = (\sqrt{1-x}, \sqrt{y}) \quad f(x, y) = (1-x, y)$$



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$$g(x, y) = \sin((x^2 - 1)^2 + y^4) \quad g(x, y) = \sin((x^2 - 1)^2 + y^4)$$

$$h(x, y) = \frac{x}{x^2 + y^2} h(x, y) = \frac{x}{x^2 + y^2}$$

$$s(x, y) = x^2 e^{y^2} \quad s(x, y) = x^2 e^{y^2}$$

Which of the following is true?

☐

Domain of  $f = \{(x, y) \mid x \leq 1, y \geq 0\}$   $f = \{(x, y) \mid x \leq 1, y \geq 0\}$

☐

Domain of  $h = \mathbb{R}^2$   $h = \mathbb{R}^2$

☐

$$g \circ f(x, y) = \sin(x^2 - 1 + y^2) \quad g \circ f(x, y) = \sin(x^2 - 1 + y^2)$$

☐

$$((g \circ f) \times h)(x, y) = \frac{\sin(x^2 + y^2)}{x^2 + y^2} \quad ((g \circ f) \times h)(x, y) = \frac{\sin(x^2 + y^2)}{x^2 + y^2}$$

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$$\lim_{(x, y) \rightarrow (0, 0)} ((g \circ f) \times h + s)(x, y) = 0 \quad \lim_{(x, y) \rightarrow (0, 0)} ((g \circ f) \times h + s)(x, y) = 0$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Domain of  $f = \{(x, y) \mid x \leq 1, y \geq 0\}$   $f = \{(x, y) \mid x \leq 1, y \geq 0\}$

$$\lim_{(x, y) \rightarrow (0, 0)} ((g \circ f) \times h + s)(x, y) = 0 \quad \lim_{(x, y) \rightarrow (0, 0)} ((g \circ f) \times h + s)(x, y) = 0$$

1 point

Match the functions of two variables in Column A with their graphs given in Column B.

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i  $\rightarrow$  4, ii  $\rightarrow$  3

☐

i  $\rightarrow$  3, ii  $\rightarrow$  4

☐

iii  $\rightarrow$  1, iv  $\rightarrow$  2

☐

iii  $\rightarrow$  2, iv  $\rightarrow$  1

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

i  $\rightarrow$  3, ii  $\rightarrow$  4

iii  $\rightarrow$  1, iv  $\rightarrow$  2

1 point

Consider two functions  $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  and  $g: \mathbb{R}^2 \rightarrow \mathbb{R}$  defined as:

$$f(x, y) = (x^2, y^2) \quad f(x, y) = (x^2, y^2)$$

and

$$g(x, y) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases} \quad g(x, y) = \begin{cases} x^2 + y^2 & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Which of the following option(s) is (are) true?

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$$g \circ f(x, y) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases} \quad g \circ f(x, y) = \begin{cases} x^2 + y^2 & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

☐

$$g \circ f(x, y) = \begin{cases} \frac{x^4 - y^4}{x^4 + y^4} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases} \quad g \circ f(x, y) = \begin{cases} x^4 + y^4 & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

☐

$\lim_{(x, y) \rightarrow (0, 0)} g(x, y)$  exists.

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$\lim_{(x, y) \rightarrow (0, 0)} g \circ f(x, y)$  exists.

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$$\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} g \circ f(x, y) = \lim_{y \rightarrow 0} \lim_{x \rightarrow 0} g \circ f(x, y) = \lim_{x \rightarrow 0} \lim_{y \rightarrow 0} g \circ f(x, y) = \lim_{x \rightarrow 0} \lim_{y \rightarrow 0} g \circ f(x, y)$$

$$g \circ f(x, y)$$

☐

$$\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} g(x, y) = 1$$

☐

$$\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} g \circ f(x, y) = -1$$

☐

$\lim_{(x, y) \rightarrow (1, 1)} g(x, y)$  does not exist.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$g \circ f(x, y) = \begin{cases} \frac{x^4 - y^4}{x^4 + y^4} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases} \quad g \circ f(x, y) = \begin{cases} x^4 + y^4 & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

$$\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} g(x, y) = 1$$

$$\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} g \circ f(x, y) = -1$$

**Numerical Answer Type (NAT)**

Consider a function  $f(x, y) = xy$ . If the directional derivative of  $f$  at point  $(a, b)$ , in direction of  $(1, -1)$  and  $(1, 1)$  are 1 and 5 respectively. Then find the value  $\frac{2a+b}{\sqrt{2}}$

2a+b.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 7

1 point

Consider the following statements:

**Statement 1:** Let  $f: R^2 \rightarrow R$  be a function such that  $\lim_{x \rightarrow a} \lim_{y \rightarrow b} f(x, y)$  exists, where  $a, b \in R$ . Then  $\lim_{(x, y) \rightarrow (a, b)} f(x, y)$  exists.

**Statement 2:** Let  $f: R^2 \rightarrow R$  be a function such that  $\lim_{y \rightarrow b} \lim_{x \rightarrow a} f(x, y) = f(a, b)$  and  $\lim_{x \rightarrow a} \lim_{y \rightarrow b} f(x, y) = f(a, b)$ , where  $a, b \in R$ . Then  $f$  is continuous.

**Statement 3:** Let  $f: R^2 \rightarrow R$  be a function such that  $\lim_{x \rightarrow a} \lim_{y \rightarrow b} f(x, y) = \lim_{y \rightarrow b} \lim_{x \rightarrow a} f(x, y) = f(a, b)$ , where  $a, b \in R$ . Then  $f$  is continuous.

**Statement 4:** Let  $f: R^2 \rightarrow R$  be a function such that  $f_x(a, b)$  exists. Then  $f$  is continuous.

**Statement 5:** Let  $f: R^2 \rightarrow R$  be a function such that  $f_y(a, b)$  exists. Then  $f$  is continuous.

How many statements are correct?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: String) 0

1 point

Consider the following two functions  $f_1, f_2: R^2 \rightarrow R$  defined as:

$$f_1(x, y) = xy + 2x, f_2(x, y) = x^2 + 2y + 3x^2y.$$

Define a matrix  $A$  as follows:

$$A = \begin{bmatrix} \frac{\partial f_1}{\partial x}(3, 0) & \frac{\partial f_2}{\partial x}(1, 1) \\ \frac{\partial f_1}{\partial y}(3, 2) & \frac{\partial f_2}{\partial y}(\sqrt{3}, 2) \end{bmatrix}. A = [\partial_x f_1(3, 0) \partial_y f_1(3, 2) \partial_x f_2(1, 1) \partial_y f_2(\sqrt{3}, 2)]$$

$\sqrt{\quad}, 2]$ .

What will be the value of  $\det(A)$ ?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) -6

1 point

### Comprehension Type Question

An earthworm is crawling about on a sunny day. We draw coordinates axes on the ground where it is crawling and find that the body temperature of the earthworm at a point  $(x, y)$  is given by the function:

$$T(x, y) = 2x^2 + 3xy + y^2$$

Use this information to answer questions 8,9, and 10.

1 point

Which of the following option(s) is (are) true?

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$$\lim_{(x,y) \rightarrow (1,1)} T(x,y) = \lim_{x \rightarrow 1} \lim_{y \rightarrow 1} T(x,y) \quad \lim_{(x,y) \rightarrow (1,1)} T(x,y) = x \rightarrow 1 \lim_{y \rightarrow 1} T(x,y)$$

☐

$T(x,y)$  is continuous at (1,1)

☐

$T(x,y)$  is a linear function.

☐

$$T(cx, cy) = c T(x, y) \quad T(cx, cy) = c T(x, y).$$

☐

If the earthworm approaches the point (1, 2) from any direction, then the body temperature of the earthworm approaches 12 units.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\lim_{(x,y) \rightarrow (1,1)} T(x,y) = \lim_{x \rightarrow 1} \lim_{y \rightarrow 1} T(x,y) \quad \lim_{(x,y) \rightarrow (1,1)} T(x,y) = x \rightarrow 1 \lim_{y \rightarrow 1} T(x,y)$$

$T(x,y)$  is continuous at (1,1)

If the earthworm approaches the point (1, 2) from any direction, then the body temperature of the earthworm approaches 12 units.

If the gradient of  $T(x,y)$  is (25, 18) at a point, then find the body temperature of the earthworm at that point.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 77

1 point

1 point

Which of the following option(s) is (are) true?

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If the earthworm moves parallel to the  $X$ -axis, then rate of change of its body temperature at the point  $(a, b)$  is given by  $4a + 3b$ .

☐

If the earthworm moves parallel to the  $Y$ -axis, then rate of change of its body temperature at the point  $(a, b)$  is given by  $2a + 3b$ .

☐

If the earthworm moves parallel to the straight line joining the origin and the point (2, 3), then rate of change of its body temperature at the point (1, 1) is given by  $\frac{29}{\sqrt{13}}$ .

☐

29.

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If the earthworm moves parallel to the straight line joining the origin and the point (0, 1), then rate of change of its body temperature at the point (1, 2) is given by 10.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If the earthworm moves parallel to the  $X$ -axis, then rate of change of its body temperature at the

point  $(a, b)$  is given by  $4a + 3b$ .

If the earthworm moves parallel to the straight line joining the origin and the point  $(2, 3)$ , then rate of change of its body temperature at the point  $(1, 1)$  is given by  $\frac{29}{\sqrt{13}}$

✓  
29.

Check Answers